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A multi-objective optimization of process parameters for laser cladding based on GA-ACO-BP and GNSGA-II

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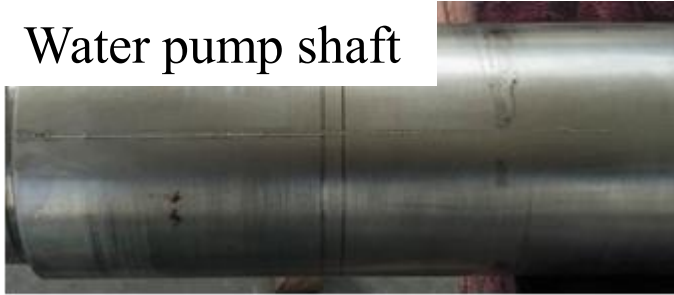
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1、Background

Water pump shaft



Crankshaft



Impeller



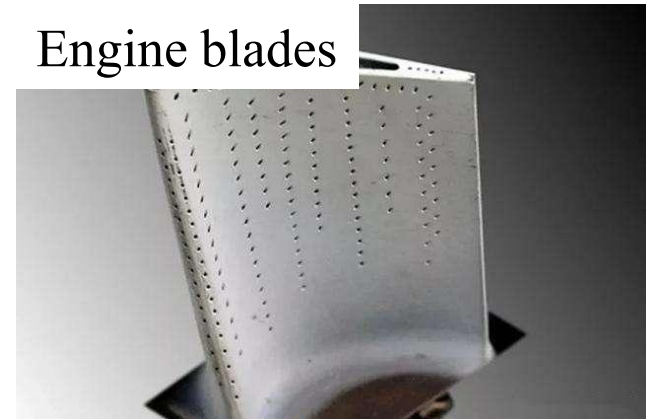
Balance Plate



Compressor Turbine



Engine blades

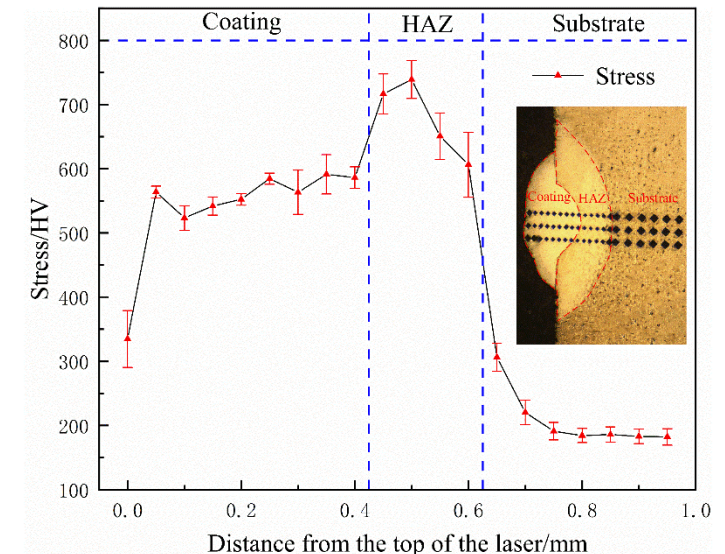
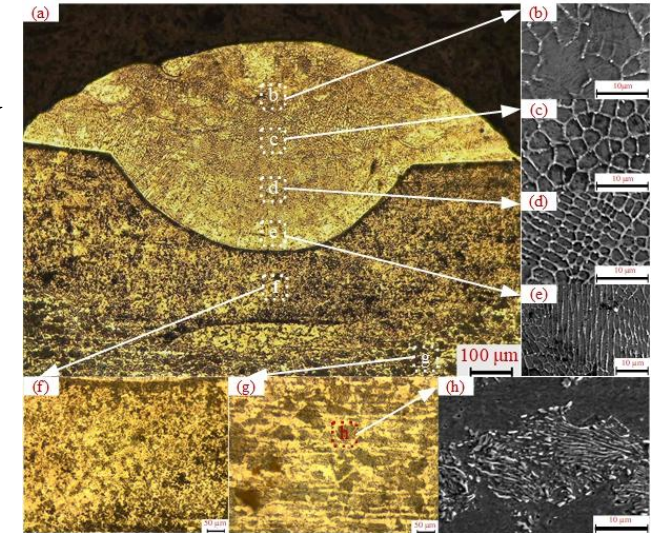
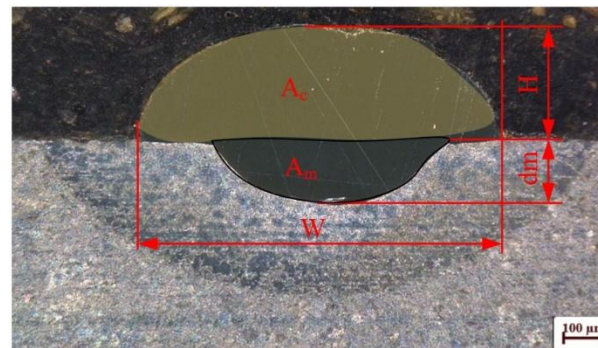
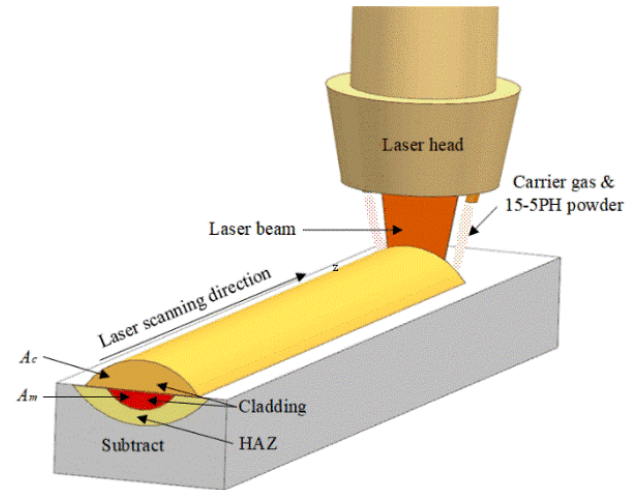
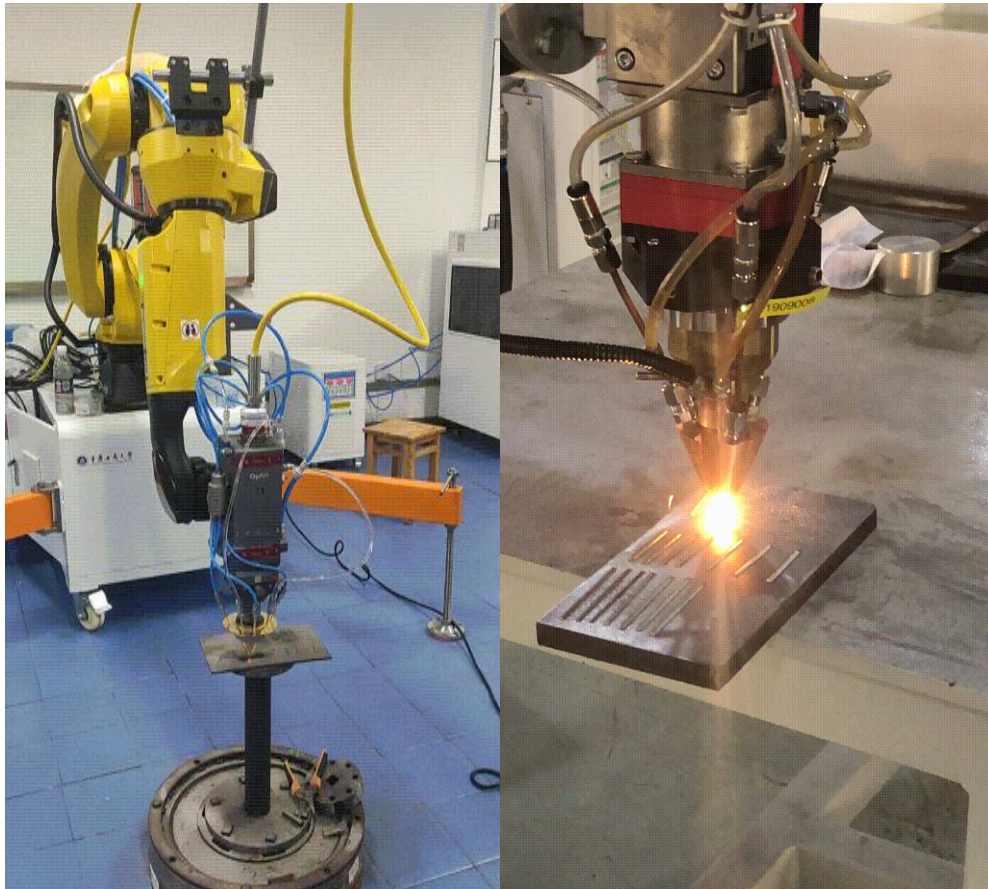


Remanufacturing of parts after failure

Surface Strengthening

1、Background

Laser surface cladding technology is the rapid heating and melting of alloy or ceramic powder with the surface of the substrate under the action of a laser beam to form a surface coating that is metallurgically bonded to the substrate material.



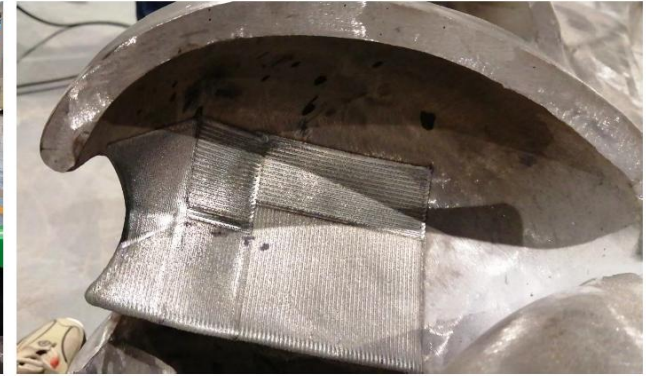
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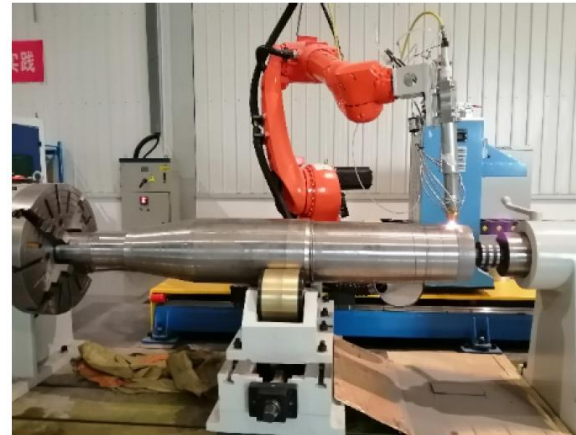
Laser cladding repair of the impeller journal part of a centrifugal pump after failure.



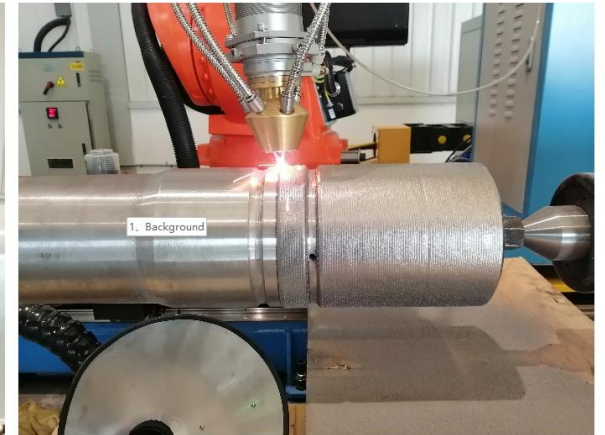
Laser cladding repair of hydro turbine unit runners.



Laser cladding repair of plungers in piston pumps.



Laser clad repair of spindles.





1、 Background

Scientific Issues

- 1、 The coupling effect of laser, powder and substrate makes the relationship between laser coating process parameters and coating quality characteristics a complex nonlinear one.
- 2、 The acquisition of optimal process parameters mainly relies on engineering experience, and the obtained optimal process parameters are inaccurate and require a lot of time and resources.

Solution method

On the basis of the established BP neural network regression prediction model, the Pareto solution set is obtained by using GNSGA-II for process parameter search with the coating quality characteristics and mechanical properties as the target, combined with engineering experience and the process range of equipment parameters as constraints. Based on the obtained Pareto solution set, the optimal portfolio of process parameters is obtained using gray correlation in combination with the VIKOR method.



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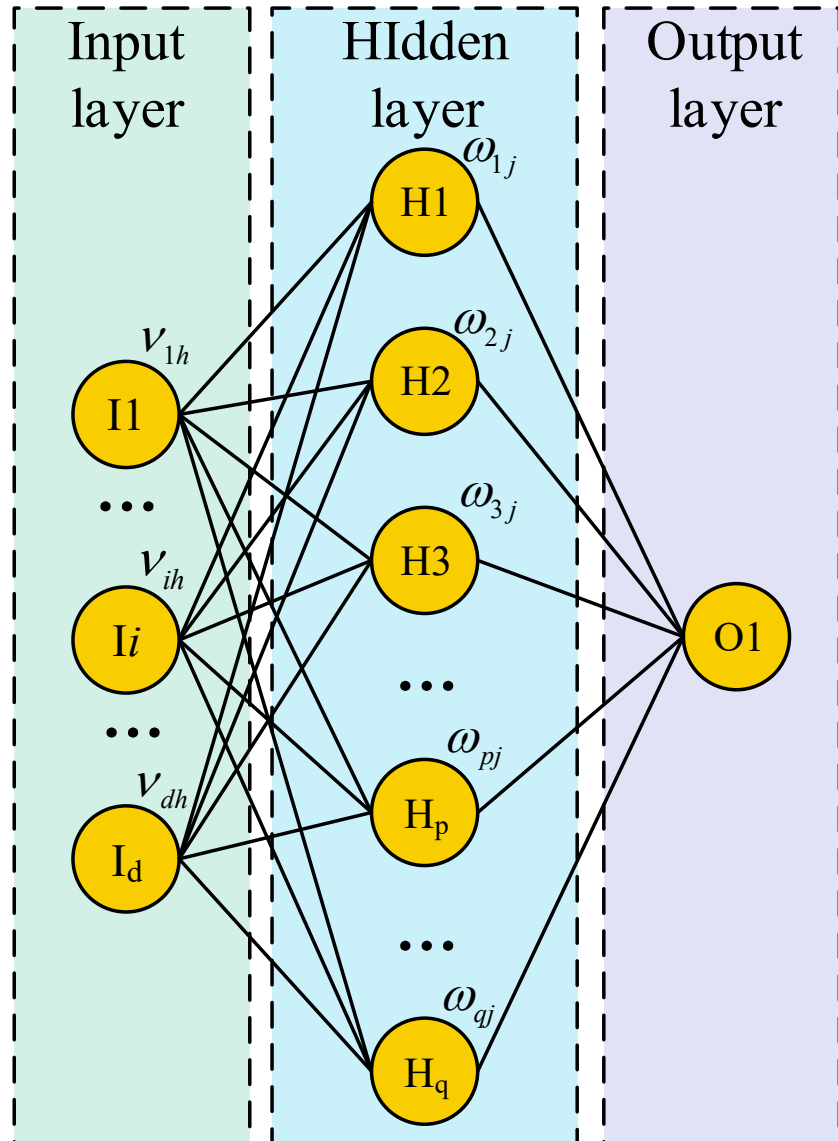
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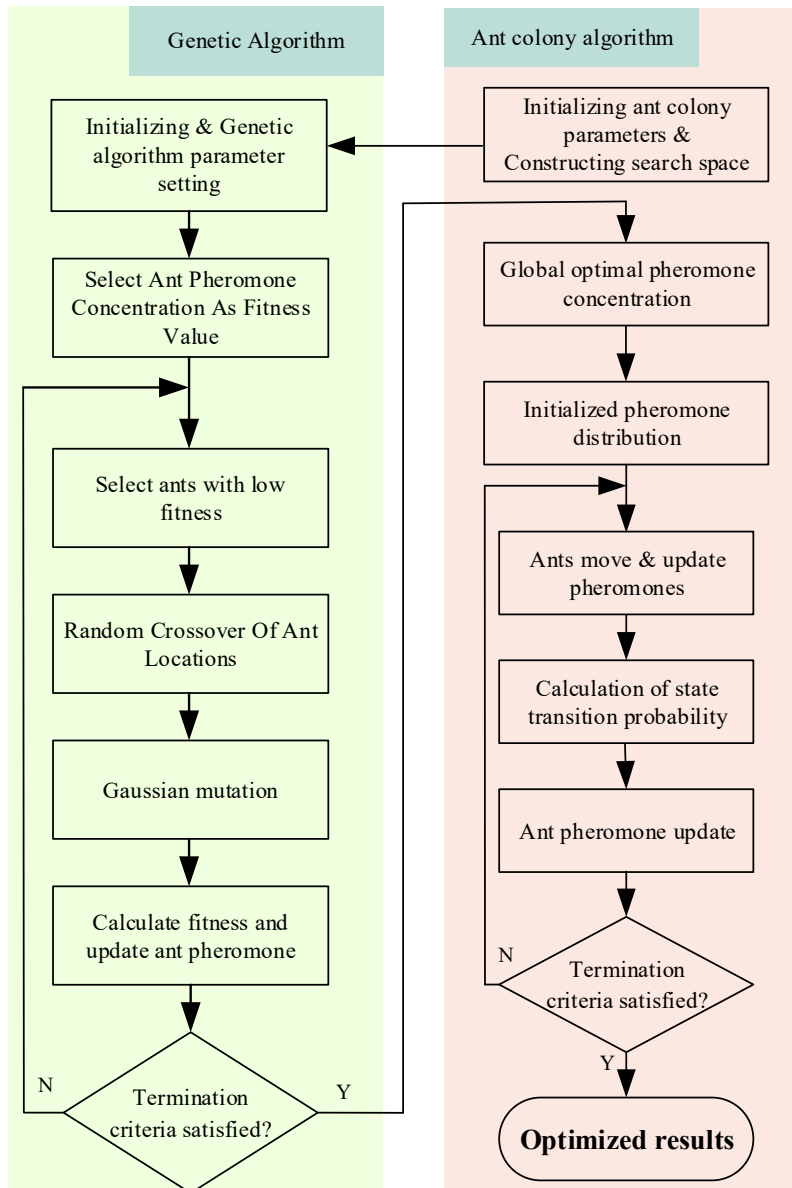


BP neural network (Back-ProPagation Network), also known as Back-Propagation Neural Network, is trained with sample data, and the network weights and thresholds are continuously modified so that the error function decreases in the negative gradient direction to approximate the desired output.

$$\left\{ \begin{array}{l} \alpha_p = \sum_{i=1}^d I_i v_{ih} \text{ The input of the } p\text{th hidden layer neuron} \\ \beta = \sum_{p=1}^q H_p \omega_{pj} \text{ Output neuron input} \end{array} \right.$$

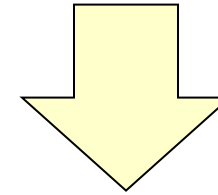
The thresholds and weights in the BP Neural Network hyperparameters determine the performance of the created network.

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The ant colony (ACO) algorithm has strong robustness and good parallel solution quality, but has the problems of slow convergence and easy to fall into local optimality.

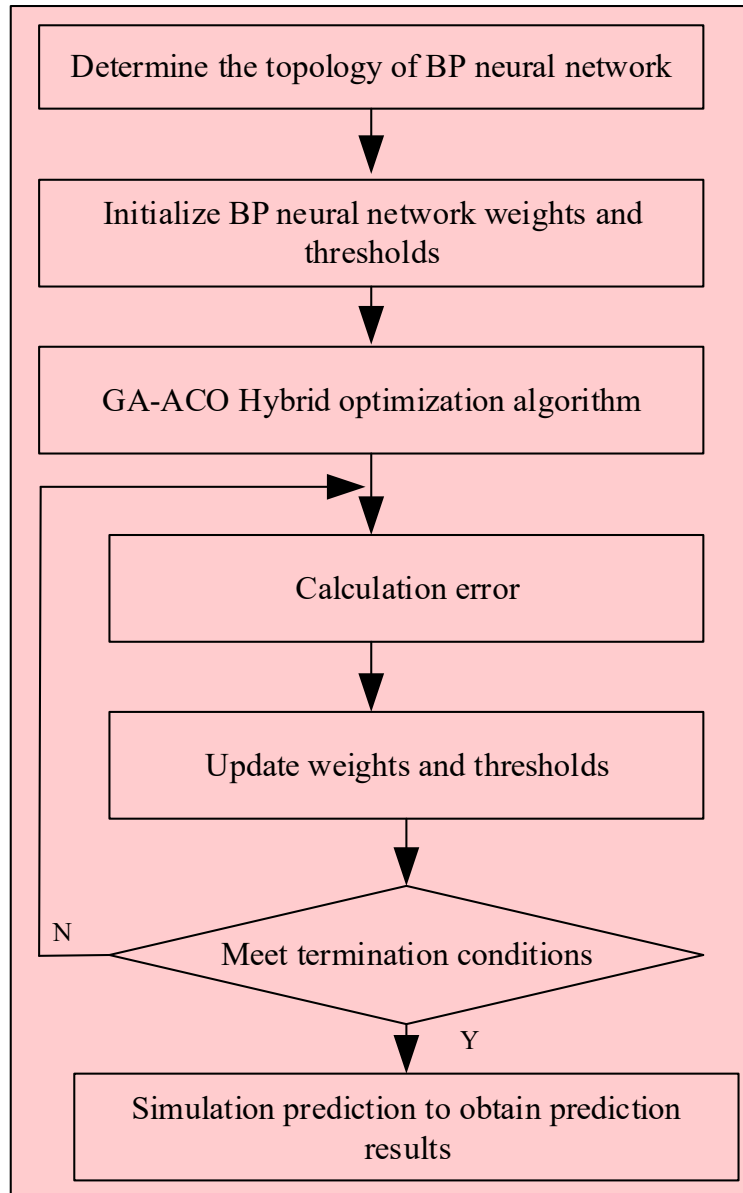
Genetic algorithm (GA) has fast and global search capability, but the solution efficiency is low.



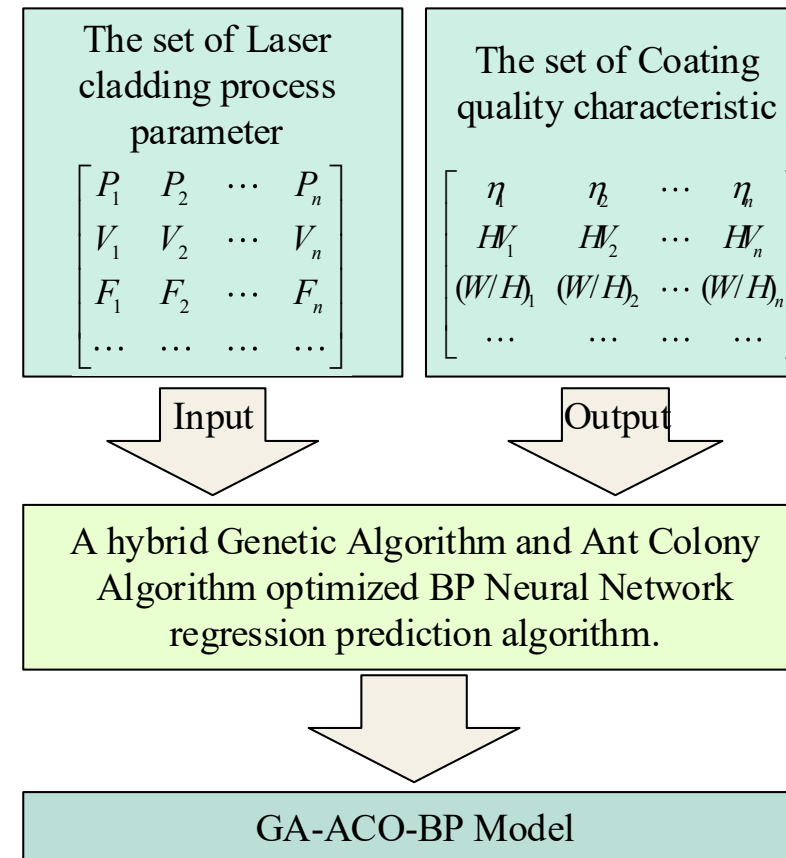
Hybrid optimization algorithm with hybrid Genetic Algorithm and Ant Colony Algorithm (GA-ACO)

In each cycle of the hybrid optimization algorithm, the GA genetic algorithm and the ACO ant colony algorithm guide each other based on updating the global optimal solution.

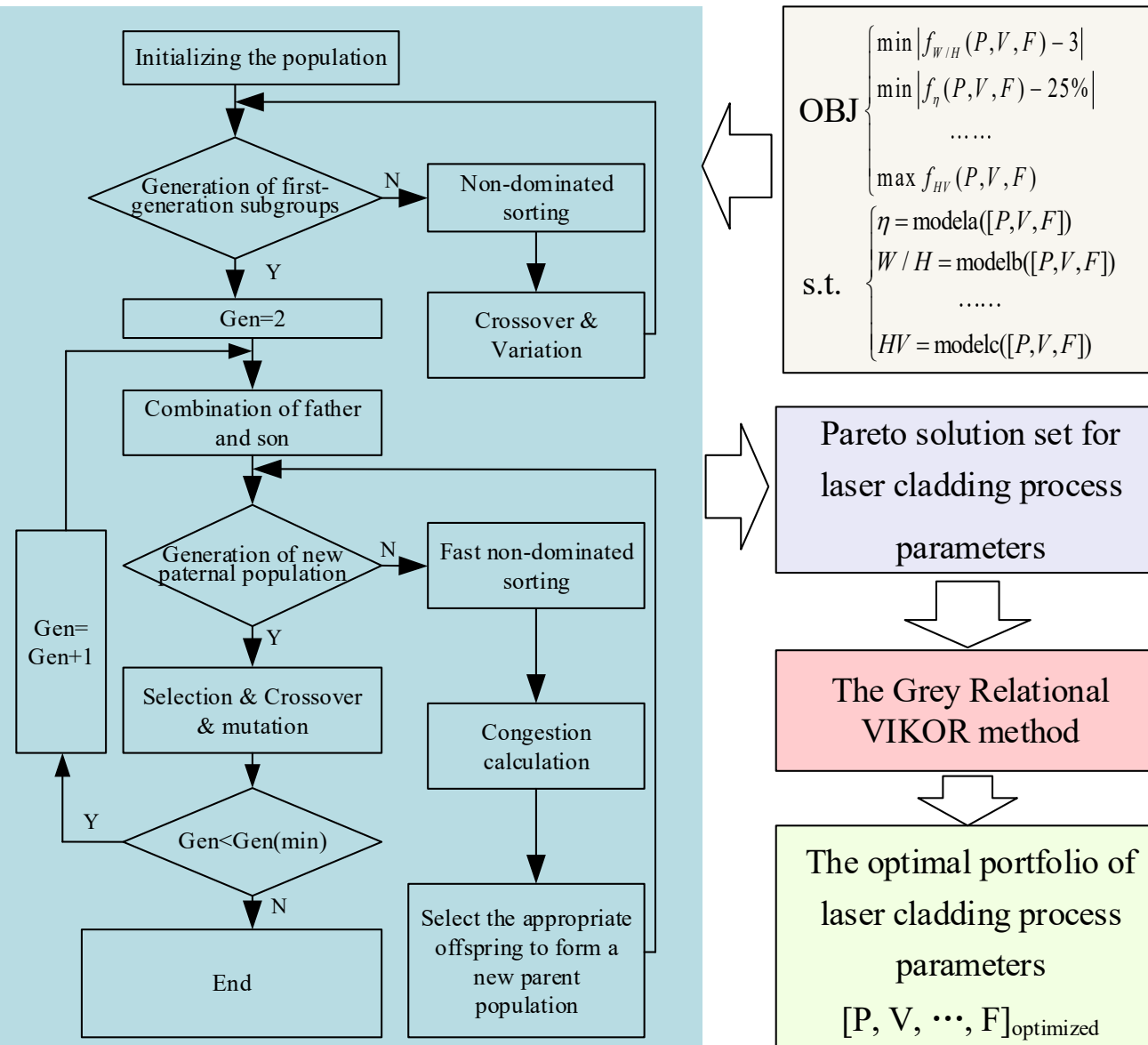
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The optimized weights and thresholds obtained from the optimization of the GA-ACO hybrid optimization algorithm are trained on the BP Neural Network.



2、 Introduction to prediction and optimization algorithms.12



To achieve optimization of laser cladding process parameters, the fast non-dominated ranking genetic algorithm with elite strategy by introducing the Gaussian distribution crossover operator (GNSGA-II) was used to optimize the process parameters of laser cladding.

The optimal portfolio of process parameters for laser cladding is obtained by the Grey Relational VIKOR method combined with the Pareto frontier solution set.



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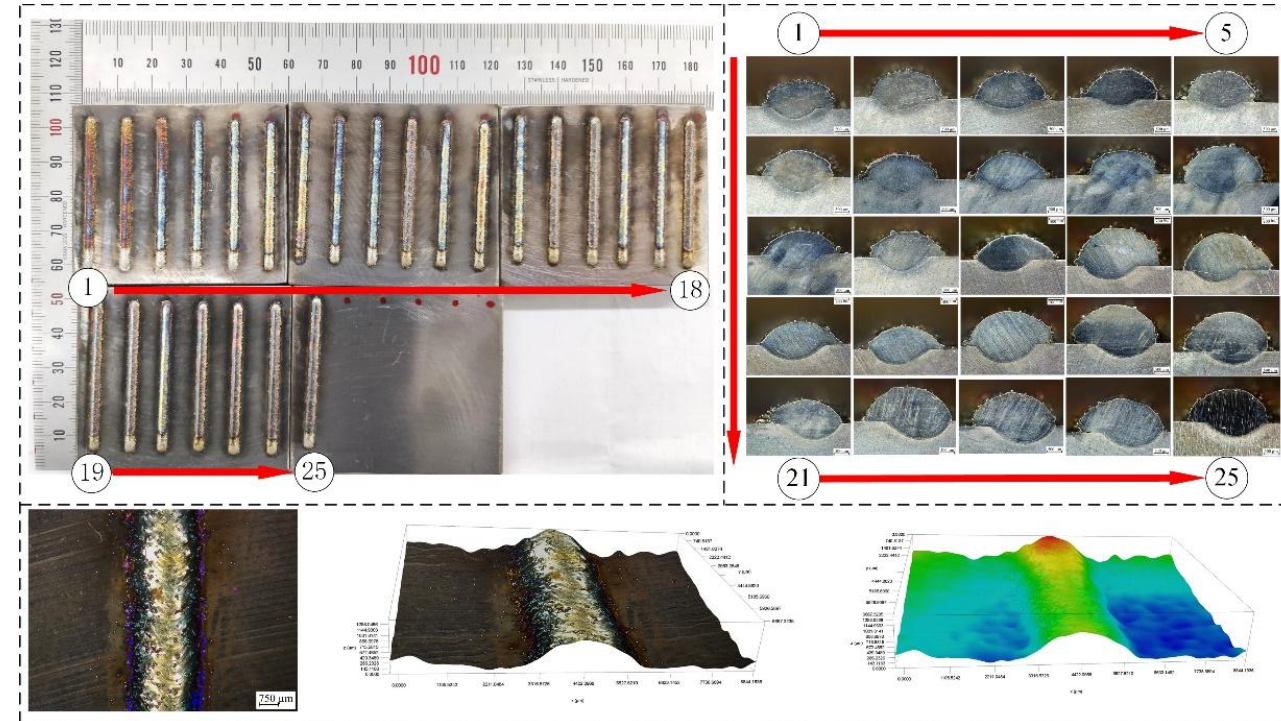
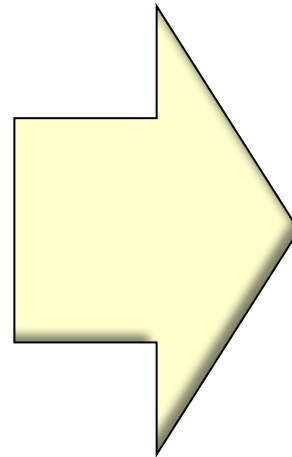
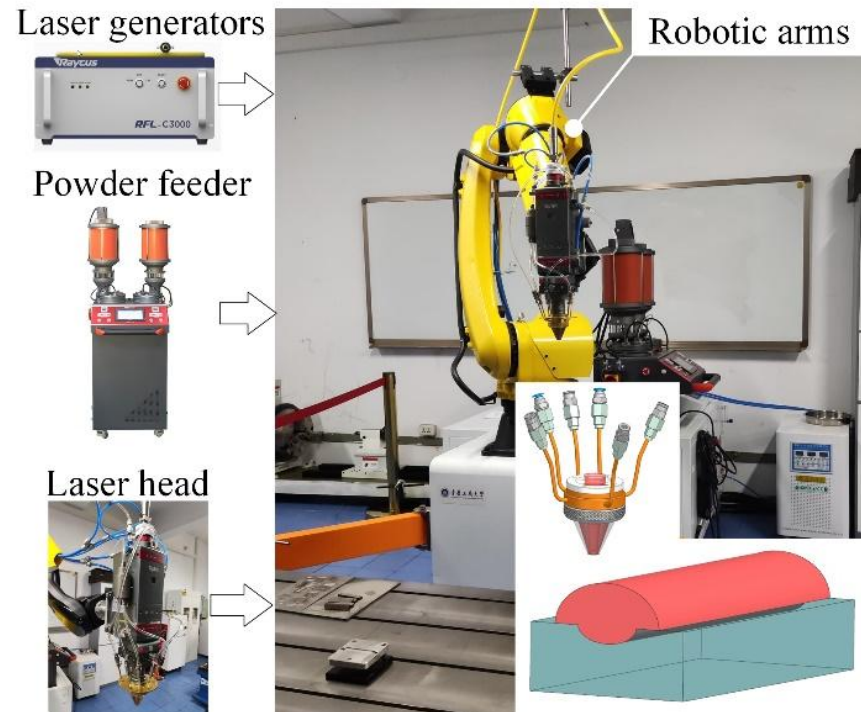
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3、Laser Cladding Experiment

To verify the proposed optimization method of laser cladding process parameters, orthogonal experiments ($L_{25}(5^3)$) were designed with laser power (P), powder feeding speed (F), and scanning speed (V) as variables, and a single-pass 15-5PH alloy cladding layer was prepared on the surface of 12Cr13 substrate using laser cladding equipment.



1、 Carbon Emissions^[1]:

$$C_E = \left\{ \left[P_{ls}(T_s - T_i) + P_{lm} \frac{60l}{V_s} \left(1 + \frac{s-d}{d(1-\alpha)} \right) N + P_{li}T_i \right] + \left[P_{ps}T_s + P_{pm} \frac{60l}{V_s} \left(1 + \frac{s-d}{d(1-\alpha)} \right) N \right] + \left[P_{ms}T_s + P_{mm} \frac{60l}{V_s} \left(1 + \frac{s-d}{d(1-\alpha)} \right) N \right] + \left[P_{cs}(T_{total} - \frac{(P_{lm}-P_{lin})T_m}{\rho v_k \Delta T c}) + P_{cm} \frac{(P_{lm}-P_{lin})T_m}{\rho v_k \Delta T c} \right] + \sum_{i=1}^n N_i P_i \frac{60l}{V_s} \left(1 + \frac{s-d}{d(1-\alpha)} \right) N \right\} C_e$$

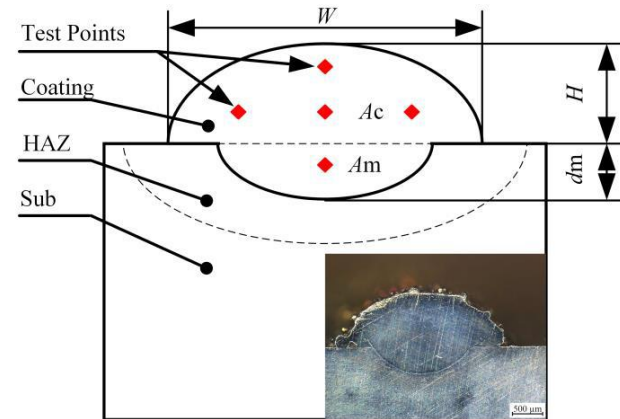
2、 Powder utilization rate^[2]:

$$\varepsilon = \frac{\rho[4(n-(n-1)\lambda)\pi R^3 + 6n(n-1)(1-\lambda)WHS + 3n\pi R^2 S]}{6n \cdot V_Q \cdot t}$$

3、 Dilution rate, aspect ratio, microhardness:

$$\eta = \frac{A_m}{A_m + A_c}$$

$$W/H = \frac{W}{H}$$



[1] JIANG Xingyu, LIU Ao, YANG Guozhe, et al. Low-carbon modeling and process parameter optimization in laser additive manufacturing process[J]. Journal of Mechanical Engineering, 2022, 58(5): 223-238.

[2] XI Wen-chao, SONG Bo-xue, WANG Zhao , et al. Forming Characteristics and Powder Effective Utilization Research of Surface Cladding Process[J]. Surface Technology, 2019, 48(3):211-218.



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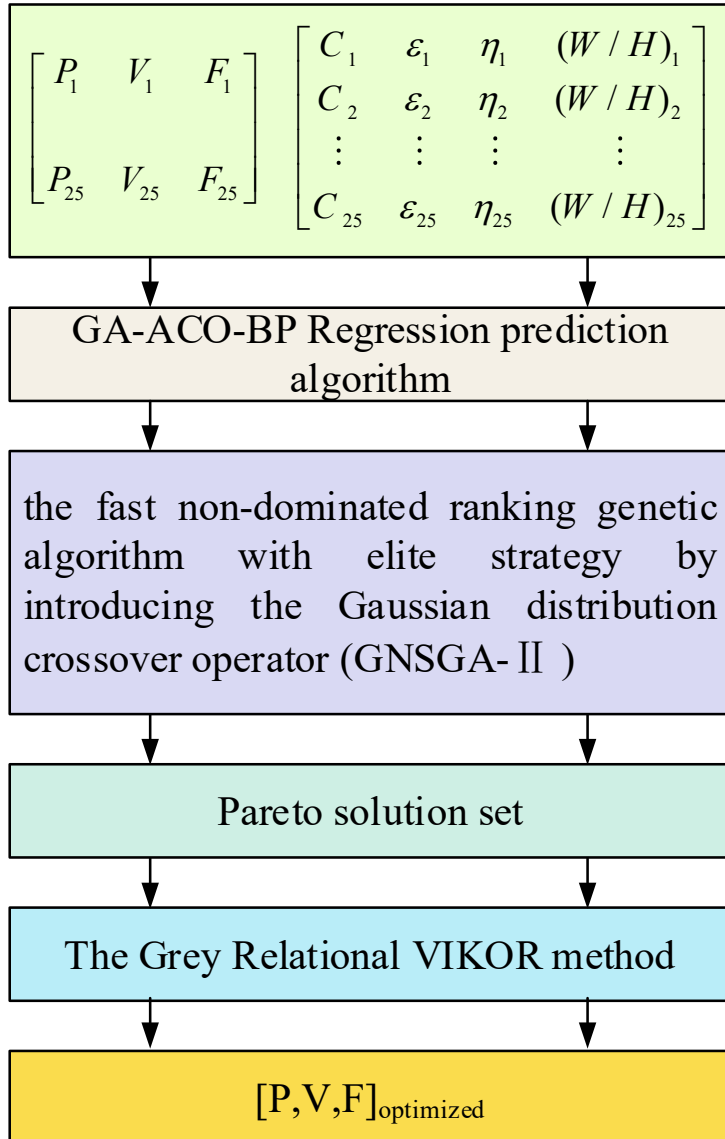
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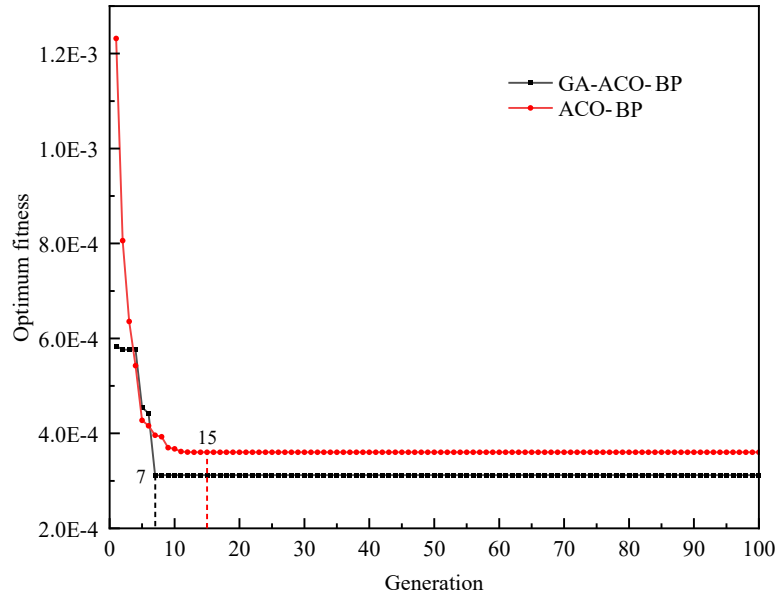
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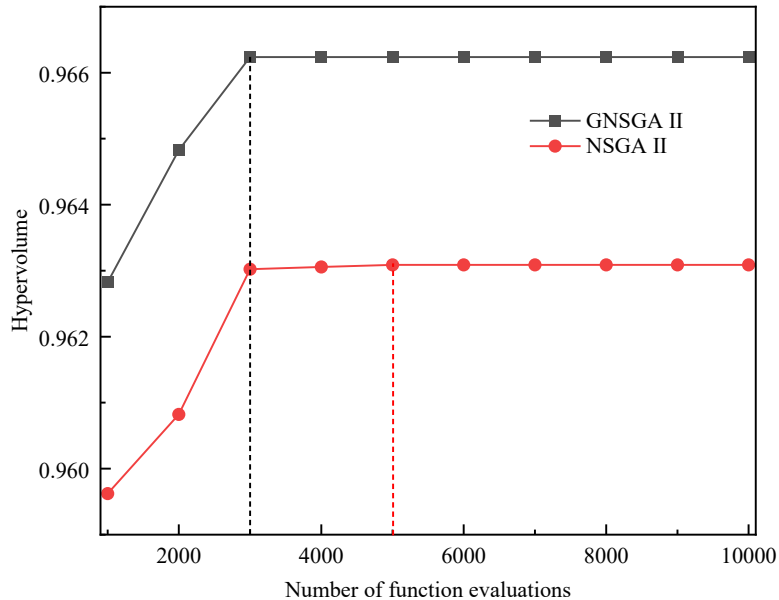
Predictive modeling between process parameters and quality characteristics is performed using the experimentally obtained data, and multi-objective optimization of process parameters is performed using the predictive model.

Through repeatability validation experiments, the results show that the developed regression prediction model can accurately respond to the complex mapping relationship between process parameters and quality characteristics.

4、 Analysis of results



By comparing and analyzing the iteration graphs of GA-ACO-BP algorithm and ACO-BP algorithm, the iteration speed and convergence speed of GA-ACO-BP algorithm are more superior. By analyzing MSE, MAPE, and R^2 of the prediction results, the fitting effect of GA-ACO-BP algorithm is obtained to be more improved than that of BP neural network algorithm.



By comparing the super volume metrics of NSGA-II optimization algorithm before and after adding Gaussian distribution operator, the results show that the GNSGA-II optimization algorithm with Gaussian distribution algorithm has better comprehensive performance



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5、 Conclusion

- ◆ The regression prediction model based on GA-ACO-BP has a good performance in predicting the coating quality characteristics.
- ◆ The Pareto solution set obtained using the GNSGA-II multi-objective optimization algorithm with the introduction of the Gaussian distribution operator provides more flexibility and convenience in the selection of process parameters for machining practice.
- ◆ The optimized laser cladding process parameters based on the Pareto solution set showed an improvement in coating quality compared to both coatings obtained from engineering experience.



5、 Conclusion

Prospect

With the development of intelligent algorithms, more and more algorithms with better performance are available, we will continue to explore more excellent intelligent algorithms for laser cladding technology.

Meanwhile, multi-track laser cladding coatings are mainly influenced by the process parameters and lap rates of single-track coatings.



Thank you !